Walnut Technologies™

AUTOMOTIVE & IOT SOLUTIONS

Smart clusters, connected devices

Smart instrument clusters with phone screen mirroring, GPS trackers and gateways, and IoT devices — designed and manufactured in India by Walnut Technologies.



Who we are

Walnut Medical & Walnut Technologies is a multi-vertical original design and equipment manufacturer (ODM/OEM). Established in 2016, we have grown from a small electronics workshop into a vertically integrated design and manufacturing partner for technologically advanced products — today including smart instrument clusters, telematics and IoT devices.

From PCB design to final assembly, every stage is in-house: electronics and firmware design, mould design, SMT, testing, quality and dispatch.

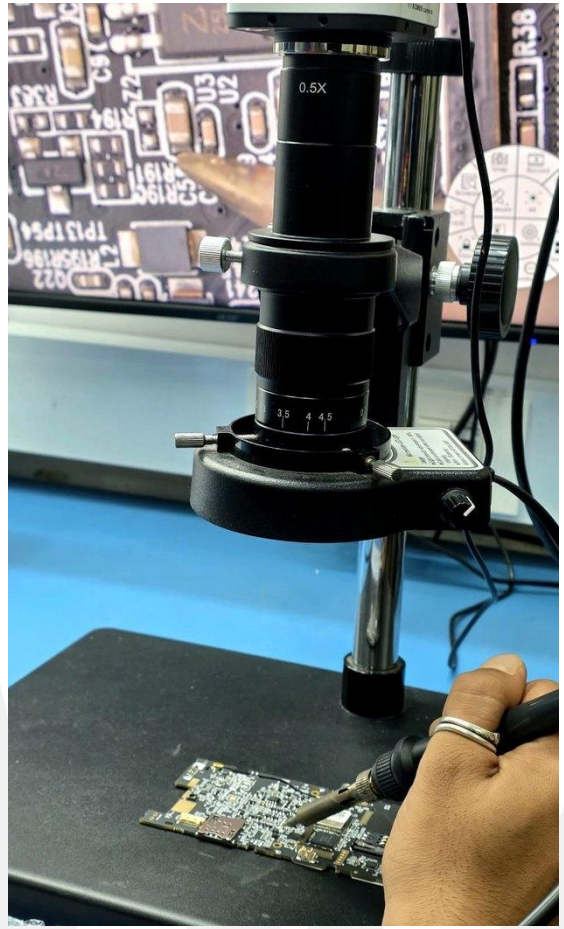
Production units

Mohali (Punjab) · AMTZ, Visakhapatnam (Andhra Pradesh) · Dhanana (Upcoming in Haryana)

Certified quality

ISO 13485 · CE · FCC · BIS · Industry 4.0 · Full traceability





SCALE & REACH

Walnut at a glance

10+
Years of excellence

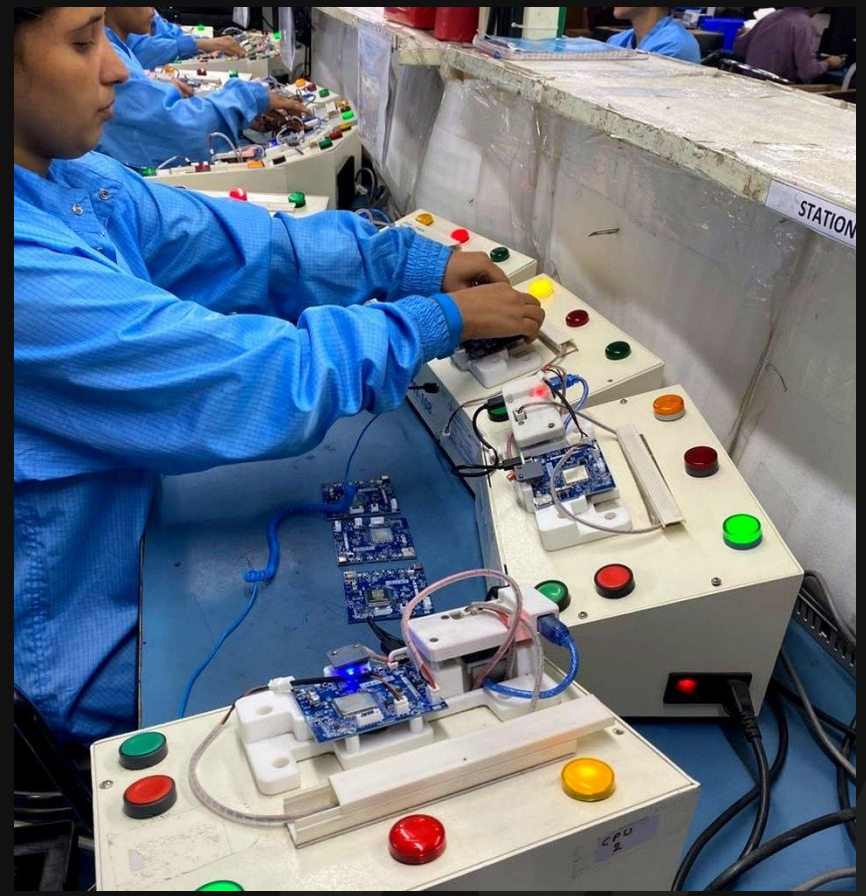
600+
Engineers & technicians

10+
Countries served

500K+
Units manufactured / month

150,000
Sq. ft facility

4
SMT production lines



Leadership

SD

Siddharth Dhawan

Director & CEO

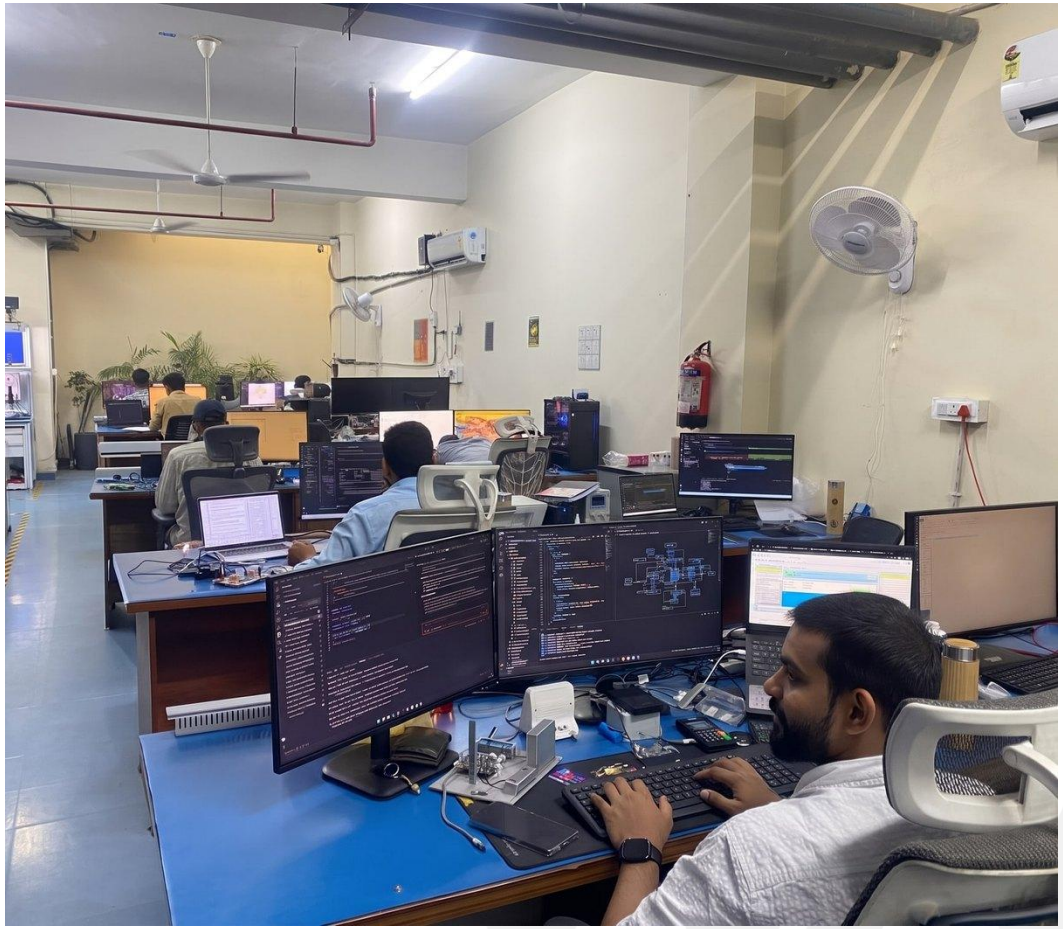
Spent 10 years at National Instruments Corp. in the USA and India, working in high-tech embedded and control systems and managing the business in India.

AB

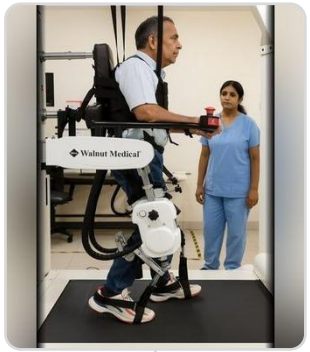
Amol Bhardwaj

Director & CTO

Leads medical device development; launched devices across diagnostics, therapeutics and robotics — EEG, ECG, rTMS and gait training robots. Expert in QC tools, SPC, APQP, PPAP, Six Sigma, Cpk and 5S; Management Representative for ISO 13485 and CE audits (2009–2014).



Our journey











2016
Walnut Medical founded

Started with a vision to make quality healthcare more accessible.

2017
Robotics & neuro-rehab

Walkex functional therapy and the WalkLab gait-training robot.

2019
Medical devices

IR thermometer and digital BP monitor, then nebulizers and oxygen concentrators.

2021
Walnut Technologies

Engineering expands beyond healthcare into electronics for every industry.

2022
Fintech · soundbox

Smart payment soundboxes and POS devices, in mass production from 2023.

2025
Smart clusters

5″ and 7″ smart clusters for 2-wheelers, tractors and vehicles.

2026
GPS tracker & IoT

GPS tracker & gateway and a growing range of connected IoT devices.

03

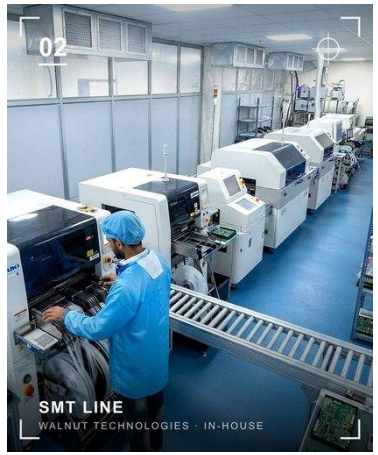
Infrastructure & manufacturing

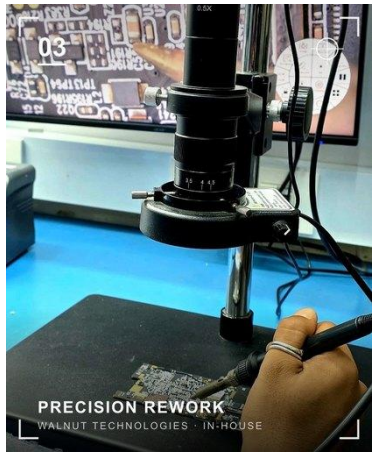
In-house electronics design, SMT, testing and quality — from prototype to 500K+ units a month.



Manufacturing infrastructure













150,000 sq. ft facility
Mohali · AMTZ · Dhanana (Upcoming in Haryana)

4 SMT production lines
Fully automatic SMT — JUKI RS-1, FUJI, 200,000 CPH

In-house test & calibration
Keysight · National Instruments · Rigol · Tektronix

PCB design to production
Design, validation, prototyping, pilot lots, full scale

IQC · LQC · OQC quality teams
Inspection at incoming, line and outgoing stages

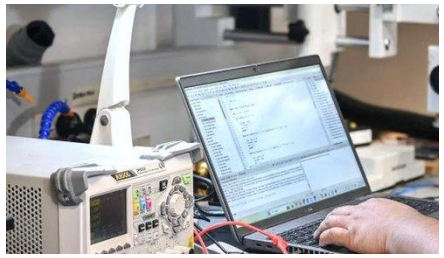
Moulds & injection moulding
Mass-production mould design and plastics in-house

End-to-End Production Process



01 Idea & Requirement

Clear direction



02 Research & Market Analysis

Right opportunity



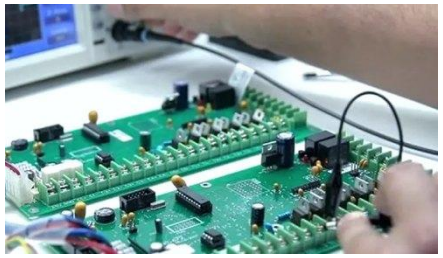
03 Design & Engineering

Optimized design



04 Prototype Development

Working prototype



05 Validation & Testing

Validated performance



06 Manufacturing Engineering

Production ready



07 Quality Assurance

Zero-defect mindset



08 Packaging & Dispatch

Safe, on-time delivery



09 After-Sales Support

Reliable support



10 Continuous Improvement

Better every day

Quality & certifications



ISO

ISO 13485

Quality management for medical devices

4.0

Industry 4.0

Connected, data-driven production with traceability

CE

CE · FCC · PCI-DSS

European, US and payment-security standards

IEC

IEC 60601

Medical electrical equipment safety

BIS

BIS certified

Bureau of Indian Standards compliance

QA

IQC · LQC · OQC

Inspection at incoming, line and outgoing stages

Compliances: ISO 13485, ISO 80601-2-69, IEC 60601-1, IEC 60601-1-2 (NABL-accredited labs), CE, FCC, RoHS

04

Automotive electronics

Smart instrument clusters in 5-inch and 7-inch, with Google Maps navigation through phone screen mirroring.



Smart instrument clusters







Next-generation digital clusters with high-resolution displays, real-time vehicle data and customisable driver interfaces — built for durability and clarity.

5-inch and 7-inch displays
Two sizes to suit compact and full-size dashboards

Android and RTOS based
Smart and cost-optimised platform options

4G · Bluetooth · Wi-Fi
Connected cluster with phone screen mirroring

BUILT FOR

- 2-wheelers
- Tractors & farm equipment
- LMV & HMV

Choose your size: 5-inch or 7-inch



5″

5-inch display

Smart instrument cluster



7″

7-inch display

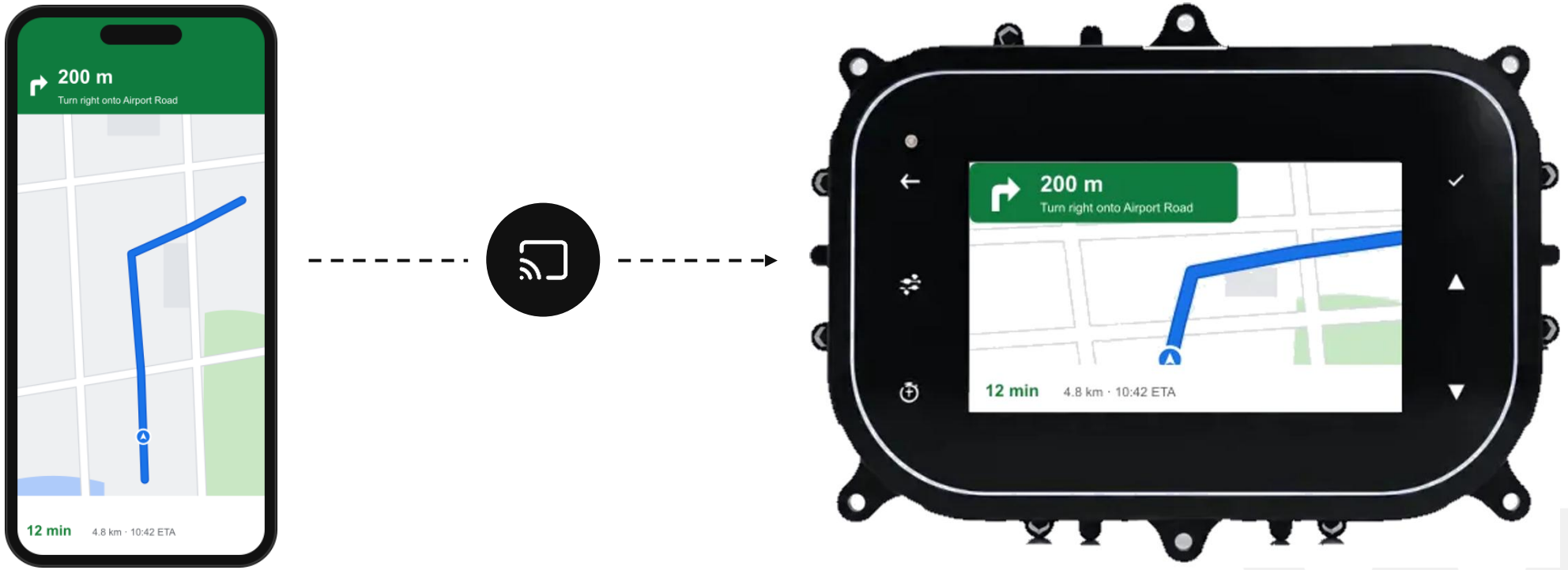
Smart instrument cluster

Configuration options

Display size	5-inch · 7-inch
Platform	RTOS · Android
Connectivity	4G · Bluetooth · Wi-Fi
Navigation	Phone screen mirroring — Google Maps on the cluster
Interface	Custom layouts, themes and brand integration
Vehicle data	Real-time speed, RPM, fuel, temperature, diagnostics

Clusters shown at relative scale. Final specifications and combinations are confirmed per project; datasheets on request.

Google maps capture on non-andriod cluster



Turn-by-turn on the dash

Google Maps navigation from the rider's phone appears on the cluster screen.

Eyes stay on the road

Directions sit in the driver’s line of sight — no phone on the handlebar or dashboard.

Works with your phone

Pair over the cluster’s wireless connectivity and start navigating.

Engineered for the road and the field

High-resolution display

Crystal-clear visuals for real-time vehicle data; anti-glare, wide viewing angles, sunlight readable.

Advanced processing

Multiple data streams processed simultaneously for instant driver feedback.

Low power consumption

Energy-efficient design for extended vehicle battery life.

Rugged build

Industrial-grade components for harsh environments.

Customisable interface

Configurable layouts and themes for different vehicles and brands.

Temperature resistant

Operating range from −40 °C to +85 °C for all climates.

Automotive engineering capabilities

01 Display technology

High-brightness LCD/TFT displays with anti-glare coating and wide viewing angles.

LCD/TFT · Anti-glare · Sunlight readable

02 Real-time data processing

Speed, RPM, fuel, temperature and diagnostics handled simultaneously.

CAN bus support · Diagnostics · Instant updates

03 Custom design & engineering

Clusters designed to your vehicle specifications, brand identity and functions.

Custom layouts · Brand integration · DFM

04 Quality & certification

Tested for temperature, vibration, humidity and electromagnetic compatibility.

IATF 16949 ready · AEC-Q100 · EMC & vibration tested

Automatic tractor electronics suite



A complete suite of smart, IoT-enabled electronics for agricultural equipment, earth-moving equipment and tractors.

Smart instrument clusters
5″ and 7″ clusters with 4G, Bluetooth and Wi-Fi

Tractor automation electronics
Electronics for automation of agricultural tractors

Telematics & tracking
GPS tracker and gateway for fleet visibility in the field

Connected platform
Cloud dashboards, mobile apps and OTA updates

05

IoT solutions

GPS tracker & gateway, 4G smart lock, connected payment devices and the cloud platform behind them.



GPS Tracker WM-MFT-4G

Smart tracking. Total control. A rugged 4G GPS tracker for real-time fleet visibility, smart geo-fencing, and route intelligence — built for tractors, auto-rickshaws, construction equipment, trucks and two-wheelers.

Precision GPS tracking
<2.5 m positioning accuracy with continuous round-the-clock vehicle tracking.

4G LTE CAT-1 connectivity
Reliable and seamless real-time data transmission anywhere on the network.

Smart geo-fencing
Customisable virtual boundaries with instant entry and exit alerts.

10,000 location history logs
Route tracking, trip analysis and complete movement records on demand.

Ignition intelligence
Real-time ACC / engine ignition status monitoring and anti-theft security.

FOTA & SOS support
Remote over-the-air firmware upgrades plus SOS emergency button.

Tractors | **Auto-rickshaws** | **Construction** | **Trucks** | **Two-wheelers**



Technical specifications — WM-MFT-4G

GNSS	GPS + GLONASS · L1 · 33 tracking channels
Accuracy	<2.5 m CEP · tracking −167 dBm · acquisition −149 dBm
TTFF (open sky)	Hot start <5 s · cold start <40 s
Cellular	LTE CAT-1 4G module · Nano-SIM
Power	9–90 V DC input · 60 mAh / 3.7 V battery · battery backup
Interface	1 ignition input · 1 output (CAN optional) · LED: GNSS blue, cellular green
Physical	82.5 × 38.5 × 15.0 mm · 50 g · built-in GSM & GNSS antennas
Environment	−20 °C to +70 °C · 5–95% RH, non-condensing
Data storage	Up to 10,000 history logs (as per customer requirement)

ADDITIONAL FEATURES

LBS positioning
Cell-tower (station mode) positioning

Geo-fence
Optional customisable boundaries

FOTA updates
Firmware-over-the-air upgrades

CAN optional
For diagnostics or data exchange

Protocol
TCP (server based) and MQTT, customisable

SOS button
Enhanced safety in critical situations

GPS gateway

Reliable GPS gateway for industrial and logistics applications — multi-sensor monitoring, real-time tracking, telemetry reporting, and fleet management with secure cloud connectivity.

200 mAh battery backup

Internal backup battery ensures continuous telemetry transmission even if main power is disconnected.

Bluetooth & GPS connectivity

Dual wireless interfaces: precise GNSS location combined with BLE for wireless sensor integration.

Multi-sensor capability

Supports Security sensor, Fuel sensor, Temperature sensor, and Humidity sensor simultaneously.

Sends multiple parameters

Transmits all telemetry parameters in real time for smarter, safer, and more efficient fleet operations.

Secure cloud gateway

Encrypted cloud communication to fleet dashboards and mobile monitoring applications.



4G smart lock



Connected smart lock for logistics, assets and infrastructure — remote access control, real-time monitoring and cloud management from anywhere.

- **Real-time status monitoring**
- **Industrial-grade IP67 design**
- **6 months standby capability**
- **4G · BLE · Wi-Fi connectivity**
- **Remote access control**
- **Cloud integration**

Connected payment devices



Smart payment soundboxes

BIS-approved soundboxes with 4G/2G, NB-IoT, Bluetooth and Wi-Fi, multi-language audio confirmation and secure integrations.

NFC tap & pay with EMV chip

Contactless and chip-insert payments — launching in 2026.

IoT management platform

SaaS platform for monitoring and managing deployed payment and IoT devices.

IoT engineering, end to end

01 Product design & development

From concept to connected product: hardware, firmware and connectivity.

Hardware · RTOS firmware · Wireless

02 Security & authentication

End-to-end encryption, secure boot and tamper detection for mission-critical devices.

Encryption · Secure element · Tamper detection

03 Manufacturing & assembly

4 SMT lines and 500K+ units per month with full traceability.

SMT & PCB · RF testing · Firmware flashing

04 Cloud & mobile platforms

Companion apps and cloud dashboards with real-time analytics.

Dashboards · Mobile apps · OTA updates

PROTOCOLS

AWS · MQTT · HTTPS · FOTA · JSON · RTOS · BLE · 4G LTE · NB-IoT

Trusted by leading brands



Healthcare

Hospitals, clinics and medical device companies

Defence & aerospace

Mission-critical electronics for defence

Banking & payments

POS terminals, soundboxes and payment devices

Pharmaceutical

Manufacturing equipment and monitoring systems

Energy & utilities

Smart metering, grid monitoring, industrial controls

Industrial & automotive

Industrial electronics, sensors and automotive

Walnut Technologies

Let's build what's next

Engineering precision with user-centric design — we welcome the opportunity to collaborate with your team.

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